

ASTRODYNAMICS I: HOMEWORK #4

Due: Wed, May 23 11:00AM, 2018

For the following problems, use $\mu_E=398,600.4418\text{km}^3/\text{sec}^2$ and $R_E=6,378\text{km}$ if necessary.

1. Prove that the eccentricity vector $\vec{e} = \frac{1}{\mu} \left(\dot{\vec{r}} \times \vec{h} - \frac{\mu}{r} \vec{r} \right)$ points to the periapsis.

Hint: use the perifocal coordinate system.

2. Rader readings determine that an object is located at

$$r = 1.2\hat{K} \text{ DU} \text{ and } \mathbf{v} = (0.4\hat{I} + 0.3\hat{K})\text{DU/TU}$$

Determine $p, e, I, \Omega, \omega, \nu, \mathcal{U}, \lambda_{true}$.

3. Given that the Mean Anomaly $M=235.4^\circ$ and $e=0.4$, find the eccentric anomaly E .
4. Given that the Mean Anomaly $M=235.4^\circ$ and $e=2.4$, find the hyperbolic anomaly H .
5. Given the coordinates $\phi_{gd} = -7^\circ 54' 23.886''$, $\lambda = 345^\circ 35' 51.000''$ and $h_{ellp}=56\text{m}$,
find the position vectors.

6. Example 3-3 in Chapter 3 of Vallado (pp. 177-178): Given $\vec{r}_{IJK} = 6,524.834\hat{I} + 6,862.875\hat{J} + 6,448.296\hat{K} \text{ km}$ on February 24, 1995, 12:00 UTC, find latitude and longitude.

7. Find the Julian Date of October 26, 1996, 2:20PM UT.
8. Find the $\text{GMST}(\Theta_{\text{GMST}})$ and $\text{LST}(\Theta_{\text{LST}})$ of August 20, 1992, 12:14PM at 104° west longitude.
9. Example 3-7 in Chapter 3 of Vallado (pp. 197-198): Find UT1 , TAI , TT , T_{UT1} , T_{TT} , T_{TDB} of May 14, 1990, 10:43 Mountain Standard Time (UTC based)